

FinnForest C1-C4 Calibration Report

Cammap / Orbcalib result compared with FinnForest ground truth

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Short conclusion

cammap/orbcalib output matches T14 direction, not T41.
Rotation error is about 0.39 degrees.
Translation is scaled, as expected for monocular SLAM.
After applying scale 2.213, translation error is about 7.1 mm.

1. What We Are Comparing

We compare the transform printed by cammap/orbcalib against the FinnForest ground truth transform named T14_opencv_style_cammap_style. The cameras are monocular, so the estimated translation can have an arbitrary scale. Rotation and translation direction can still be meaningfully compared.

```
Estimated cammap/orbcalib
  euler: -0.0750192  -56.0893      0.21423
  trans: -0.261505   0.00426842  -0.136738

Ground truth: T14_opencv_style_cammap_style
  euler:  -0.602037826  -55.909690436   0.836021159
  trans_m: -0.581736657   0.006461918  -0.296923530
```

2. Direction: T14, Not T41

The estimate has the same sign pattern and orientation direction as T14. If it were T41, it would follow the inverse transform direction. So the correct comparison for this result is against T14_opencv_style_cammap_style.

```
Estimated translation direction: -x, +y, -z
GT T14 translation direction:    -x, +y, -z

Conclusion: cammap/orbcalib output ~= scaled T14
```

3. Rotation Result

The Euler differences are small, but Euler angles are only a component-wise view. The cleaner single number is the relative rotation angle between the two rotation matrices.

```
Euler difference = estimated - ground truth
```

```
roll:   +0.527018626 deg  
pitch:  -0.179609564 deg  
yaw:    -0.621791159 deg
```

```
True rotation error:  
0.391523 deg
```

Rotation interpretation

The orientation estimate is very close to the FinnForest ground truth.

4. Raw Translation Result

Before correcting monocular scale, the translation magnitude looks too small. This is expected: monocular SLAM cannot know metric scale from images alone.

```
Raw translation vectors
```

```
estimated: -0.261505    0.00426842  -0.136738  
GT T14:    -0.581736657 0.006461918 -0.296923530
```

```
Vector lengths
```

```
|estimated| = 0.295128  
|GT|        = 0.653164
```

```
Raw residual norm without scale correction:  
0.358068 m
```

5. Monocular Scale Correction

To compare a monocular result against metric ground truth, we allow one global scale factor on the estimated translation. Rotation is not scaled. Only the translation vector is multiplied by a single number.

```
scaled_estimated_translation = s * estimated_translation
```

```
Best least-squares scale:
```

```
s = dot(estimated, GT) / dot(estimated, estimated)  
s = 2.213025814
```

```
Length-ratio check:
```

```
|GT| / |estimated| = 2.213156281
```

This means the cammap/orbcalib translation is about 0.4518 of the ground-truth metric scale. Equivalently, multiply the estimated translation by about 2.213 to compare it in meters.

6. After Applying Scale

```
Scaled estimated translation
-0.57871732  0.00944612  -0.30260472

Ground-truth T14 translation
-0.581736657  0.006461918  -0.296923530

Scaled residual
0.00301934  0.00298421  -0.00568119

Scaled translation error norm:
0.007092 m  ~=  7.1 mm
```

Translation interpretation

The translation direction is correct.
The translation magnitude differs because of monocular scale.
After one scale correction, the translation is very close to ground truth.

7. Final Interpretation

This calibration result is strong. It should be read as a monocular-scale estimate of T14. The rotation is close, the translation direction is close, and the metric translation agrees well after applying the single global scale factor.

```
Final compact result
Direction:                T14
Rotation error:           0.391523 deg
Monocular scale factor:   2.213025814
Translation direction error: 0.622134 deg
Scaled translation error:  0.007092 m
```